

Stock Synthesis Application to Black Sea Bass

Gavin Fay, Lisa Chong, Jason McNamee,
Edward Camp, Sam Truesdell, Kiersten Curti

Outline

- Previous SS application
- Description of model approach
- Model specifications
- Diagnostics
- Results

Previous Stock Synthesis application

SS models developed for SAW62 (Fay & Cadrin 2016, Fay et al. 2016)

- Investigated spatial, seasonal, and sex-specific dynamics
- Large number of fleets and surveys, fit to length composition, age-at-length data
- Fit to tag recapture data
- Included hermaphroditism

Simulation testing to evaluate estimation performance & model structure assumptions

- Models without spatial differences in recruitment patterns did not perform well
- Including seasonal movement was less critical than capturing recruitment

For this Research Track, updated the previous SS application, data, & decisions

Modeling Approach for this Research Track

Bringing to up to date version of SS

Aligned some assumptions with those of WHAM (more on next few slides) for comparison purposes

Incorporated WG feedback to take advantage of SS features and emphasizing not using data that would require double-using / borrowing, etc.

Model configuration

- 2 regions (north and south)
- 2 six-month seasons (Jan-Jun [1], Jul-Dec [2])
- Model years: 1989-2021
- Ages: 0-15+
- 2cm length bins, up to 70cm
- 2 “growth patterns” (stock components)
 - Northern recruits
 - Southern recruits
- 8 Fishing fleets
 - Commercial, North and South, season 1 and 2
 - Recreational, North and South, season 1 and 2

Biology and life history

- Biology

- Natural mortality fixed at 0.4 for both N & S, and for males and females
- Separate growth curve parameters for N & S, and by sex
 - Some sharing of growth parameters among sexes within a growth pattern
- Decadal blocks in
 - Weight at length
 - Maturity at length
- 2014-2022 change in Length at age parameters in Northern recruits (VBK & Linf)
- Hermaphroditism modeled, males included in SSB, asymptotic rate of sex change estimated

- Movement

- End of season 2 (fall), an estimated % of Northern recruited fish in the North move South
- End of season 1 (spring), estimated % of Northern recruited fish in the South move North
- No movement/mixing of Southern recruited fish

Stock-recruit dynamics

- Single 'global' stock recruit relationship
- SRR Beverton-Holt, with steepness (h) fixed at 0.95
- Recruits (age 0s) happen at start of season 2
- Expected recruitment linked to environment time series of bottom temperature
 - Input as standardized z-scores
 - Lagged so that winter bottom temperature in year y corresponds to age 0s in year $y-1$.
 - Coefficient of link to environmental index estimated
- Annual recruitment deviations 1987-2019
- SigmaR fixed at 0.7
- Time-varying proportion of total recruits to North
 - Estimated as random walk over time, 1990-2019

Initial conditions

- Stock assumed to be in a fished equilibrium in start year 1989
 - with 1987, 1988, 1989 recruitments estimated
- Initial F_s estimated for each of the eight fleets, fit to average catch 1989-1993.

Fishing fleet configuration & data

Aggregated commercial fleets (combined trawl & non-trawl)

Recreational removals input in numbers

Selectivity and retention assumed as logistic functions of length. Forced fall fleets to not catch age 0s.

15 % discard mortality rate incorporated in selectivity/retention specification for recreational fleets

- Fit to time series of catches, discards (alive & dead for recreational), length compositions (retained & discarded), age-at-length compositions
- Removed length comps in years/fleet/seasons that would require borrowing
 - e.g. when sampling incomplete across market categories
- Removed small number of observations of <8cm fish in the spring in occasional years
- Used Discard CVs from regional estimates (gives time-varying estimate)
- Selectivity blocks
 - N Comm spr: 1989-1997 selectivity & retention, 2014-2019 retention asymptote
 - S Comm spr: 1989-1997 selectivity & retention, retention asymptote changes in trawl-only discard yrs
 - N Comm fall: 1989-1997 selectivity & retention, 2014-2019 retention asymptote
 - S Comm fall: 1989-1997 selectivity & retention, retention asymptote changes in trawl-only discard yrs
 - N Rec spr: 1989-2000, 2001-2008, 2009-2021 selectivity & retention
 - S Rec spr: 1989-2000, 2001-2008, 2009-2021 selectivity & retention
 - N Rec fall: 1989-2000, 2001-2008, 2009-2021 selectivity & retention
 - S Rec fall: 1989-2000, 2001-2008, 2009-2021 selectivity & retention

Index data

Matched WHAM model in terms of index selection

Sensitivity analysis looked at fitting to disaggregated indices

Indices

- VAST - Spring North and South (index & age composition)
 - Asymptotic selectivity
- RecCPA - fall N & S (index, selectivity same as rec fleet [which includes all discards])
 - fixed variance inflation added, 0.15 in addition to the annual standardization model-based CVs

Ages & Age-at-length data

- Don't want to double use age data
- Tried to fit VAST model to length comps, but unsuccessful
- Only fit to spring North & South VAST age comps that used 'preferred ALKs'
- All other data then available to fit as age-at-length
 - Fishery-based age-at-length data that was used in spring VAST not used
- Captures spirit of intent to not double-count **AND** also use the VAST indices

Tag recapture data

- Fit to data available from tag recapture results as described in Moser & Shepherd (2009)
- No update to data except to aggregate recaptures by fleet to match aggregated commercial fleets
- For each tag release group, SS fits to
 - Number of recaptures over time (negative binomial)
 - Proportion of recaptures by fleet (multinomial)
- Overdispersion set to 2 for all Tag release groups
- Reporting rate & Initial Tag loss fixed per estimates in Moser & Shepherd 2009
- Not fitting to recaptures made in same time period as release
- Release groups with few (<10) recaptures removed from likelihood

Composition Data (length, age, & age-at-length)

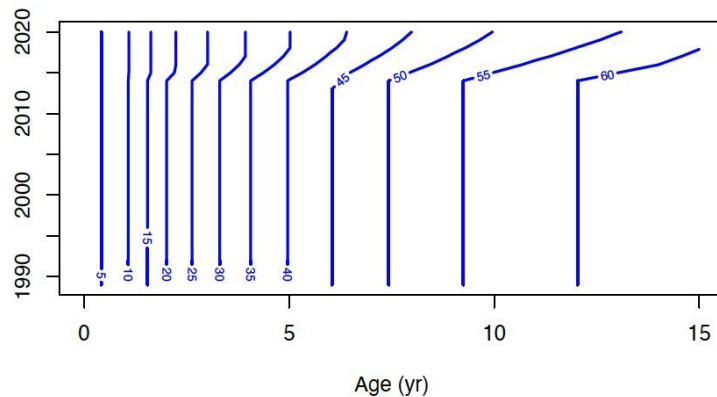
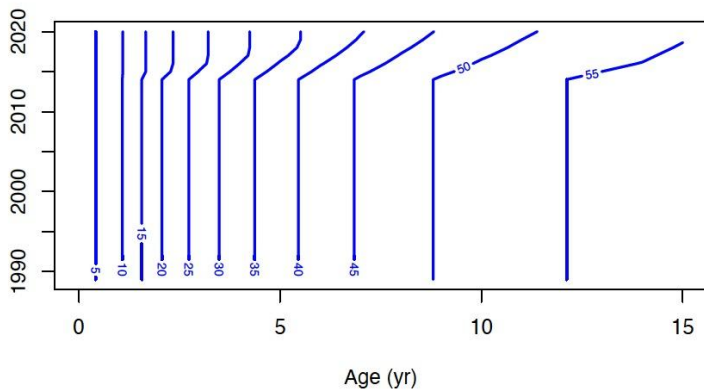
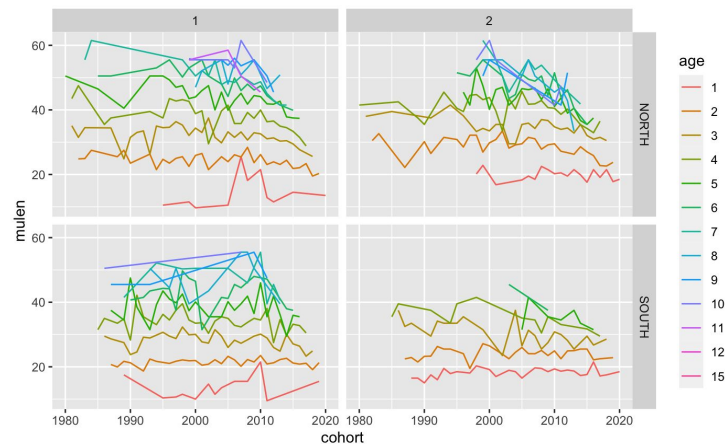
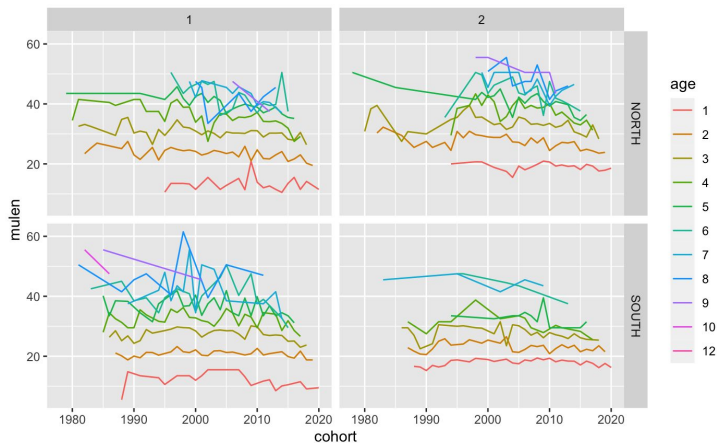
- Multinomial for everything
- Approximation of McAllister-Ianelli method for reweighting
 - 1 or 2 iterations of weighting to have approximate harmonic mean $\text{effN} \approx \text{adjusted input N}$
 - Inspection of mean lengths/weights index per Francis method
- Commercial discard length compositions were assigned to separate 'fleets' to allow for different weightings for both retained and discarded length data
 - Opens door for self-weighting likelihoods but this not explored here

Diagnostics & fit summary

- Index data
 - North indices fit well
 - South not able to fully match increase in late 90s
- Discard
 - 2014-2019 block in North helps to capture increased discards of 2011 year class
 - Adding years of changed retention improves fit to discards in South
- Lengths
 - Fits generally OK, some residual patterns remain
- VAST Age Comps
 - Fits generally OK, some cohort patterning in residuals - e.g. age comp apparent strength of 1999 and 2011 year classes not fully accounted for.
- Age at length
 - regional & time varying growth helps fit these data, some remaining evidence for mis-specification
- Tag recaptures
 - Overall number of recaptures in first couple years since release fit well, over-estimated in later years. Distribution of recaptures among fleets less well captured.

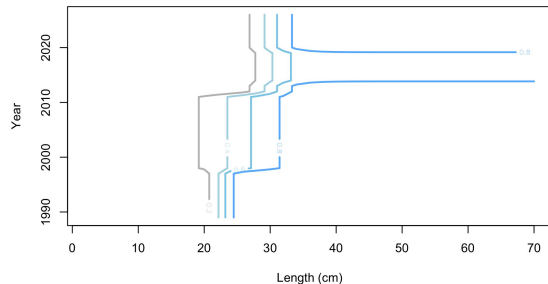
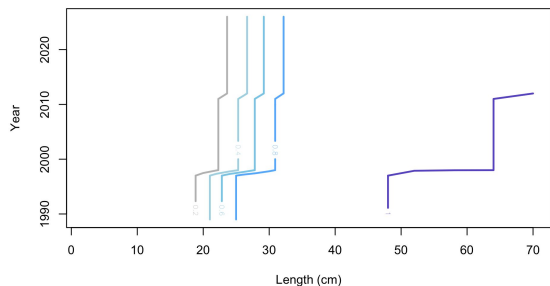
Growth

Cohort-specific
size at age (data)

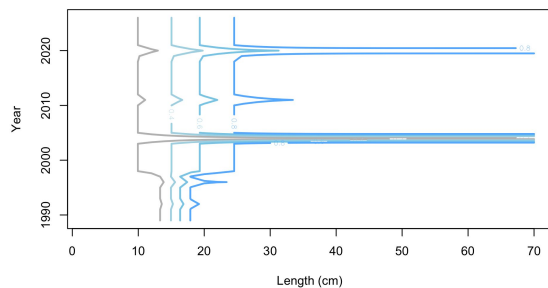
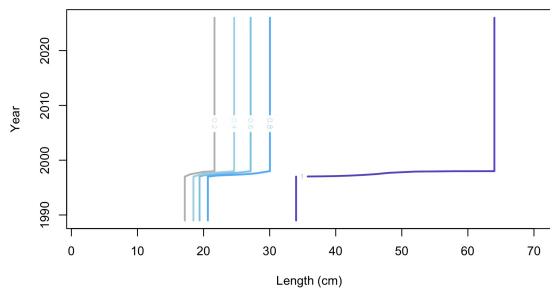


Estimated time varying growth (northern recruits)
Left females, Right: males

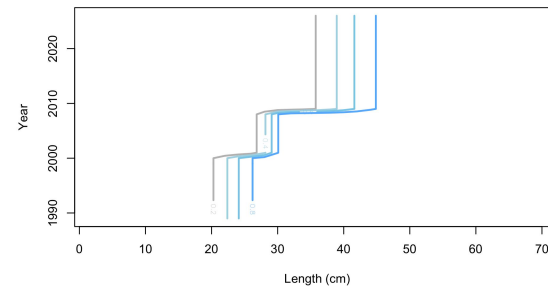
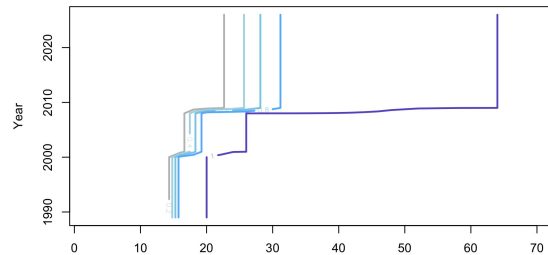
Example estimated selectivity & retention



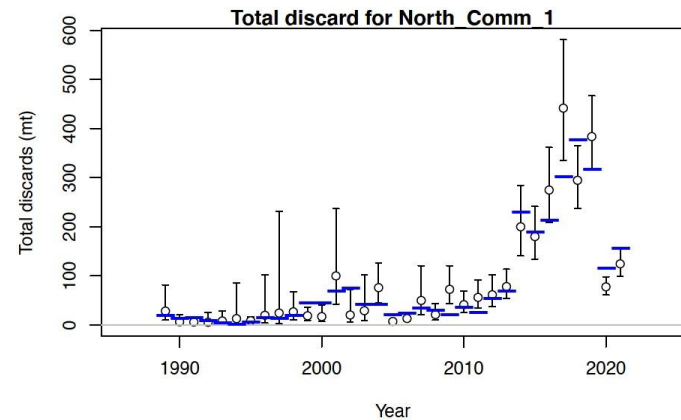
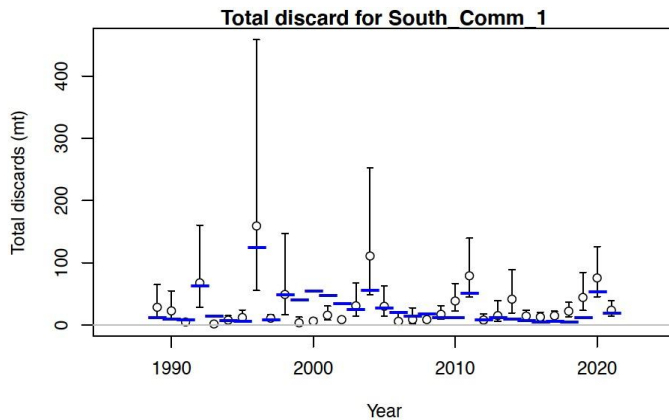
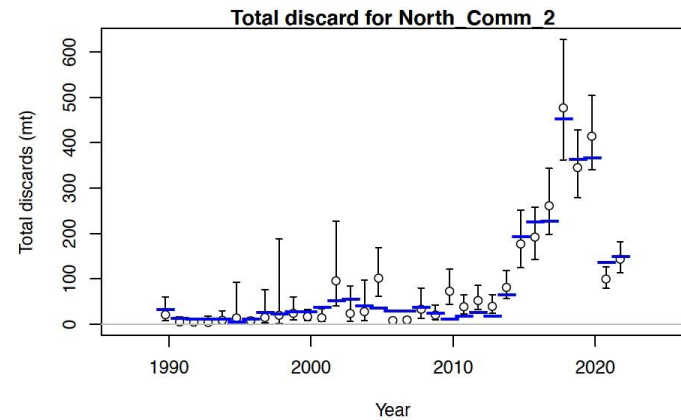
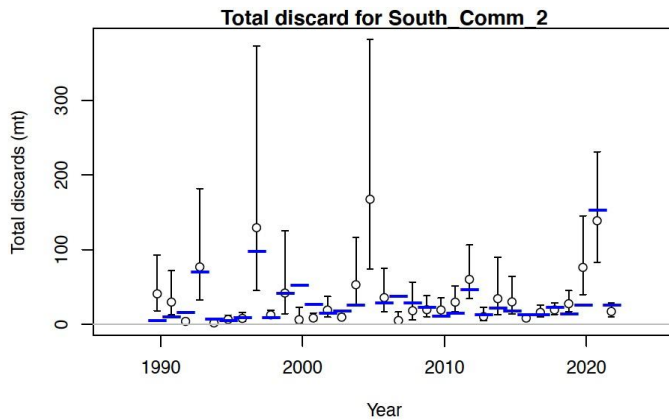
Selectivity and retention at length, N Comm Spring



Selectivity and retention at length, S Comm Fall

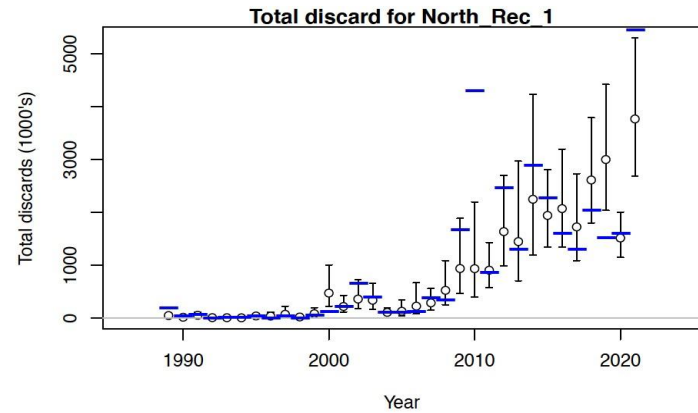
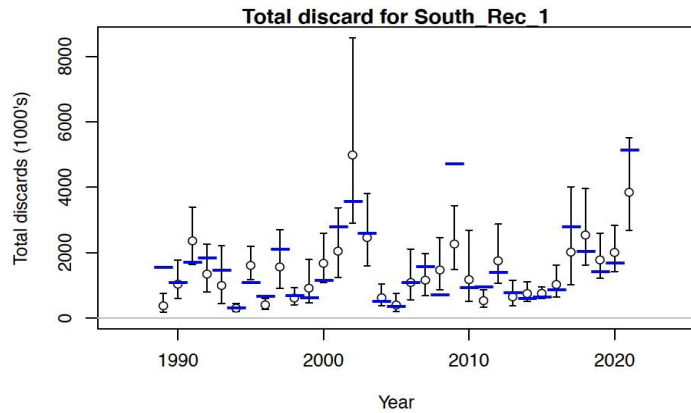
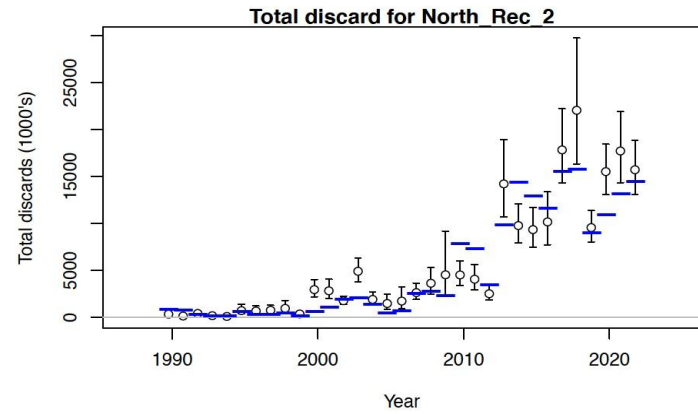
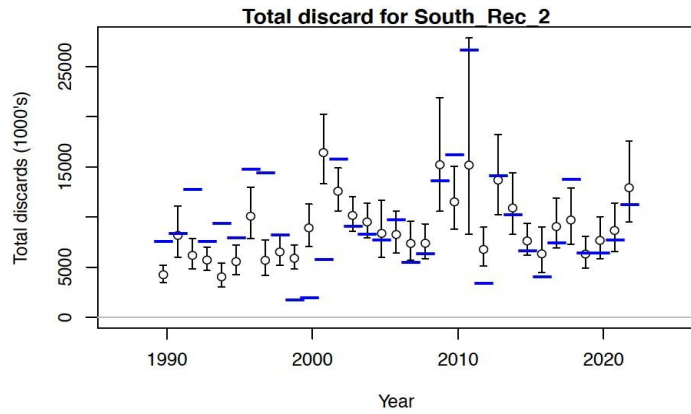


Selectivity & retention at length, N Rec Spring



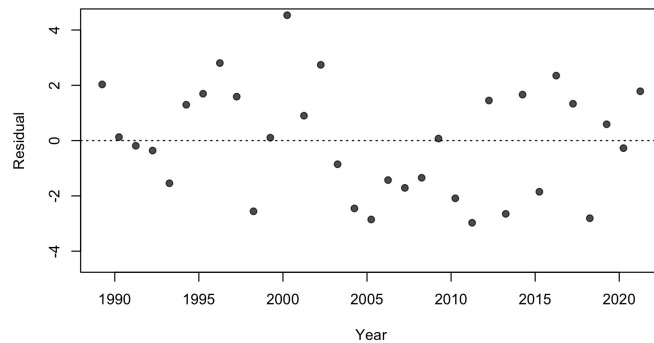
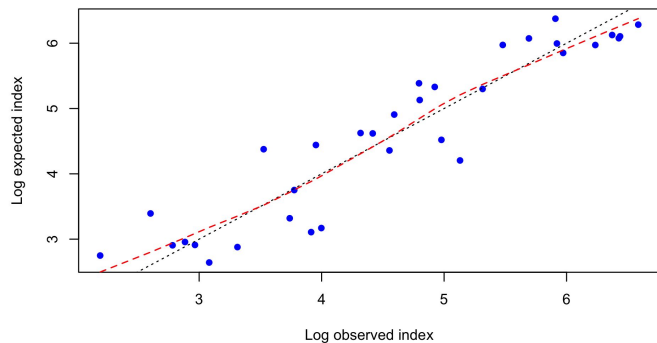
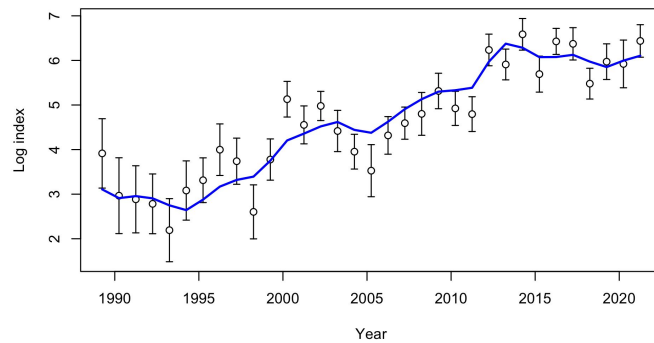
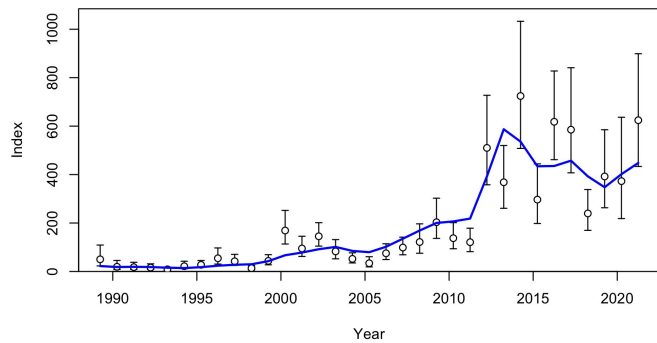
Discards

Fits to Commercial discard estimates. Season 1 = Spring, Season 2 = Fall

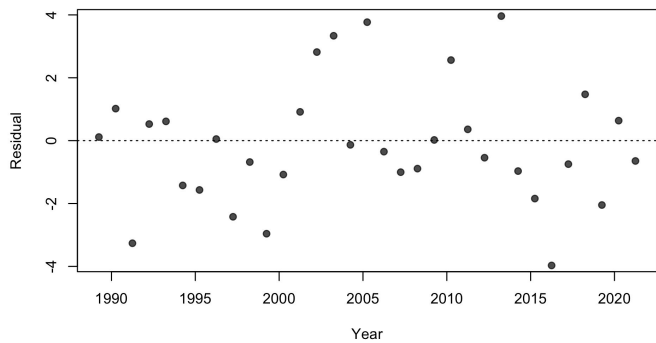
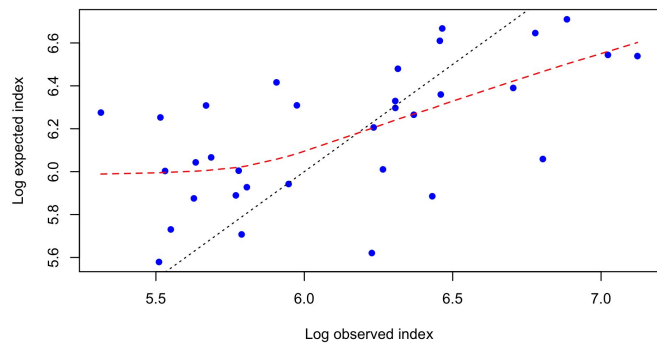
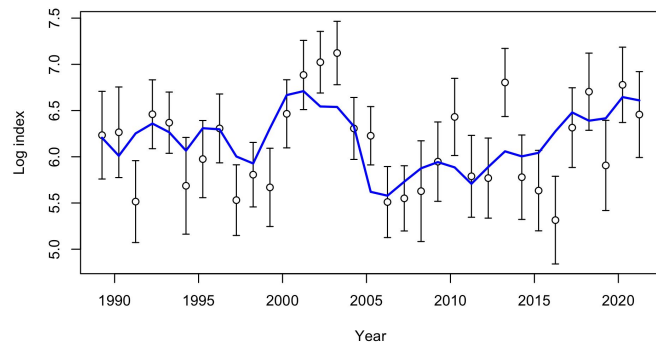
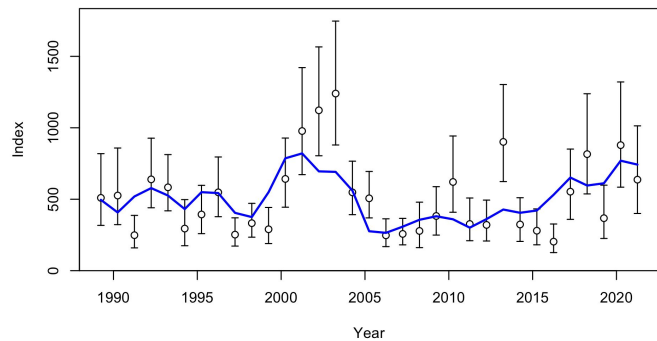


Discards

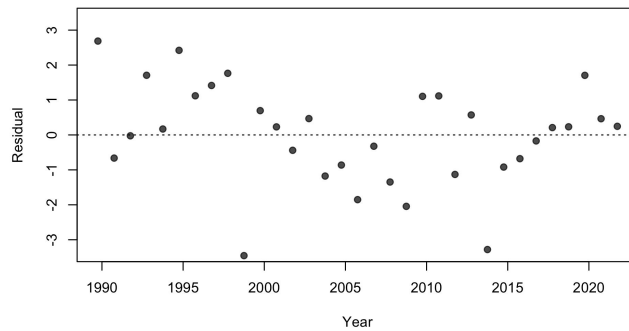
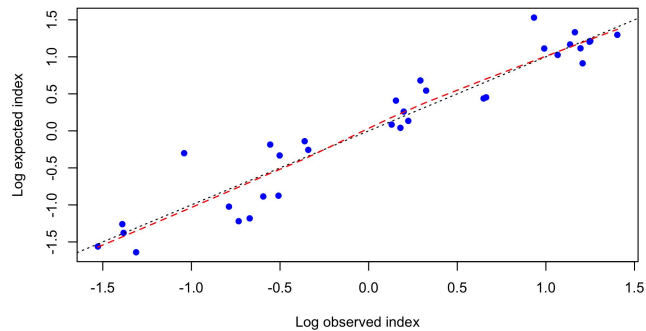
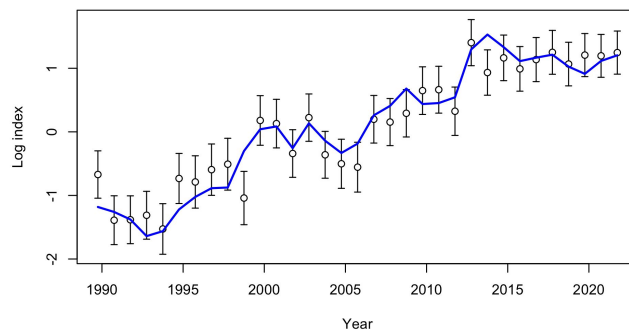
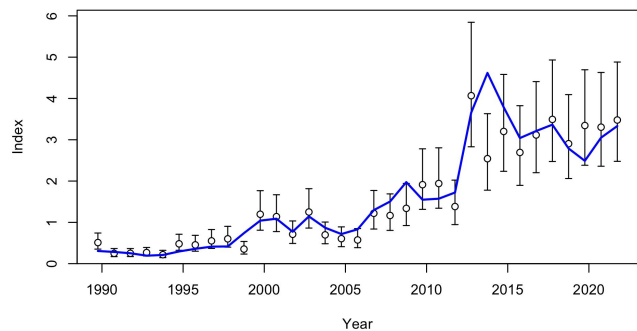
Fits to Recreational discard estimates. Season 1 = Spring, Season 2 = Fall



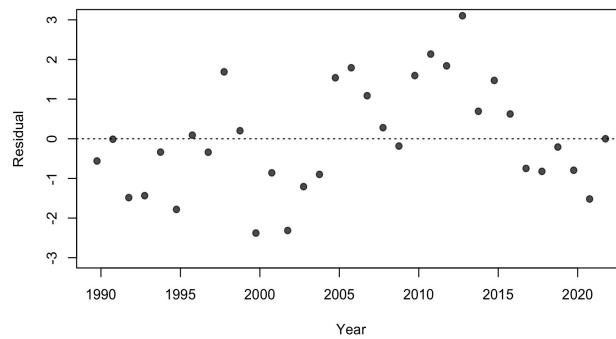
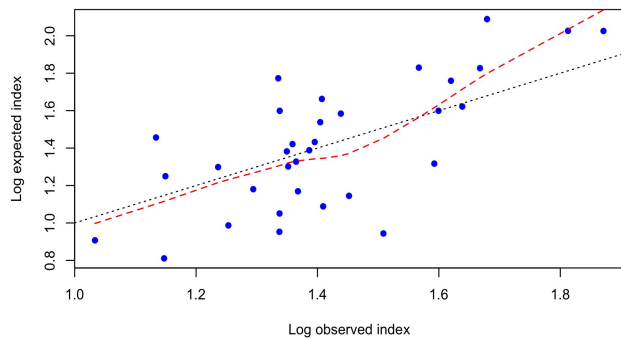
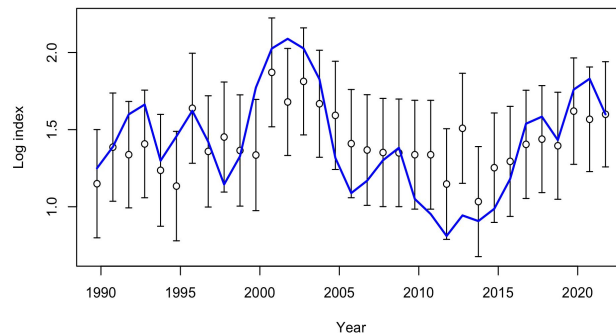
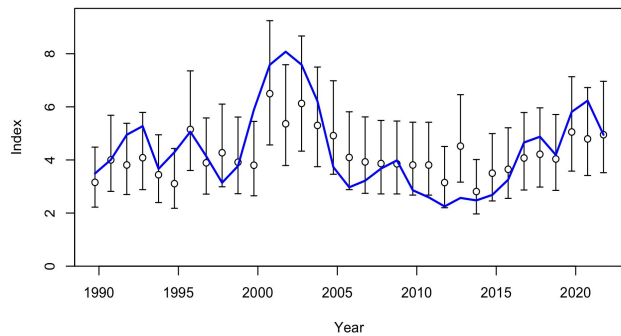
Fit to Spring North VAST index



South Spring VAST index



Fit to North RecCPA

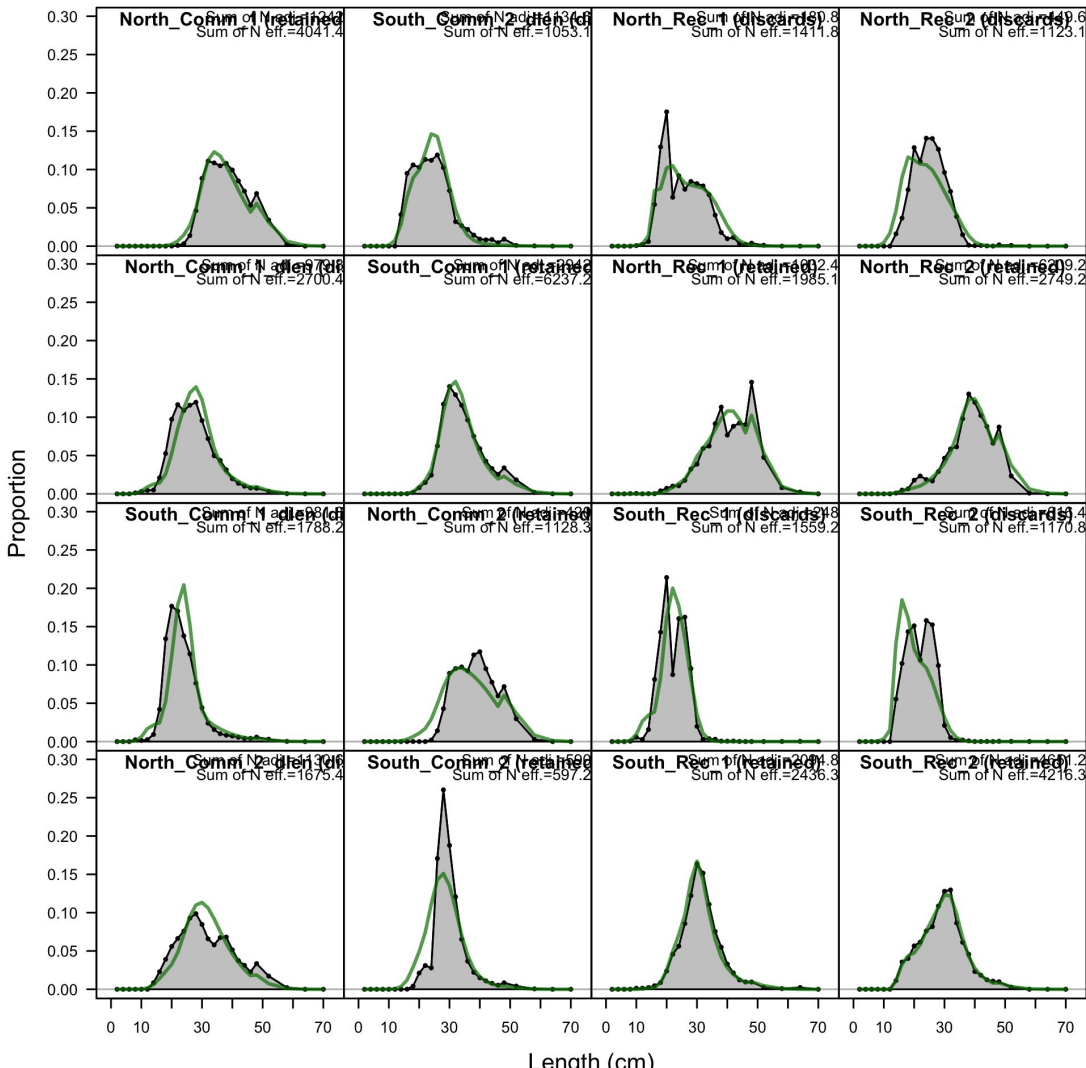


Fit to South RecCPA

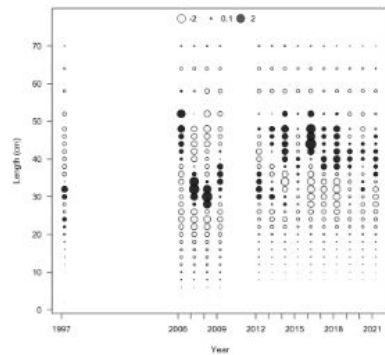
Length Composition

Fit to length compositions aggregated over time

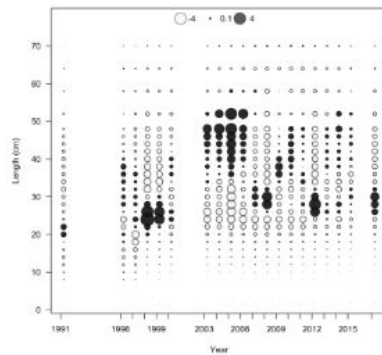
- Patterns in residuals do remain
- As with WHAM, evidence for gradual change in selectivity over time in recreational data rather than step changes (as modeled via blocks)



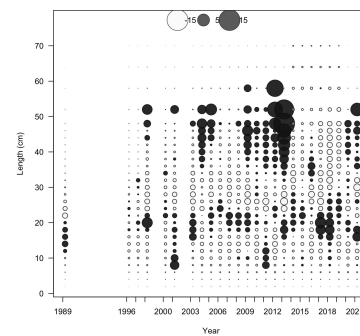
(a) Retained North Commercial Spring



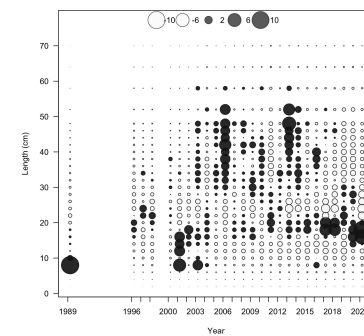
(b) Retained South Commercial Spring



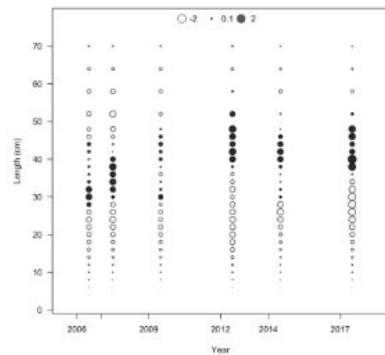
Discard North Commercial Spring



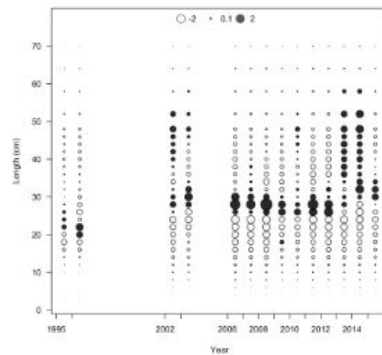
Discard South Commercial Spring



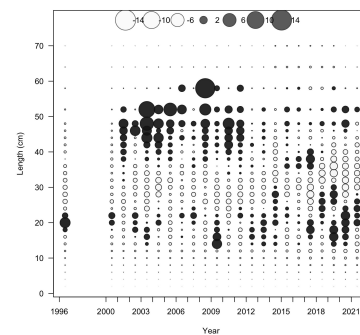
(c) Retained North Commercial Fall



(d) Retained South Commercial Fall



Discard North Commercial Fall



Discard South Commercial Fall

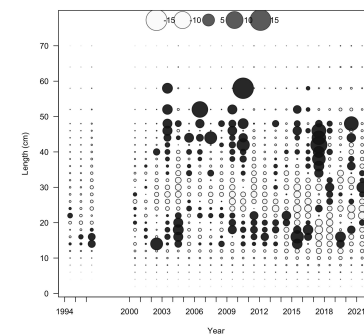
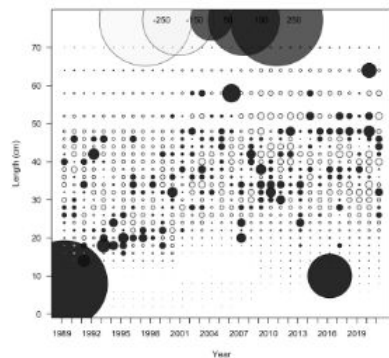
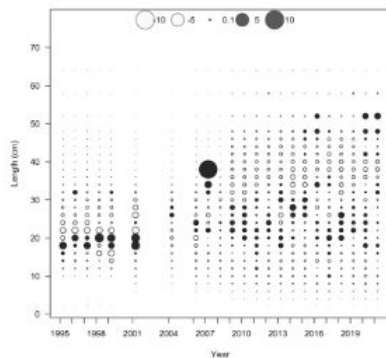


Figure 14: Pearson residuals for fits to commercial retained length composition data

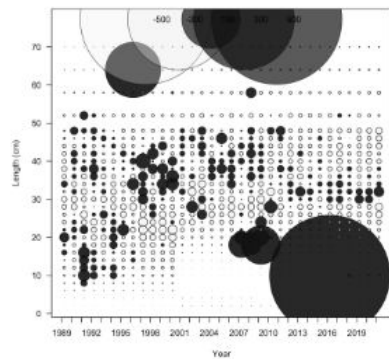
(a) Retained, N Rec Spring



(b) Discards, N Rec Spring



(c) Retained, S Rec Spring



(d) Discards, S Rec Spring

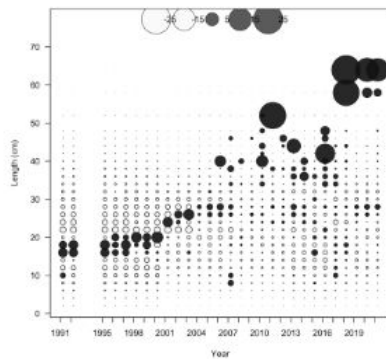
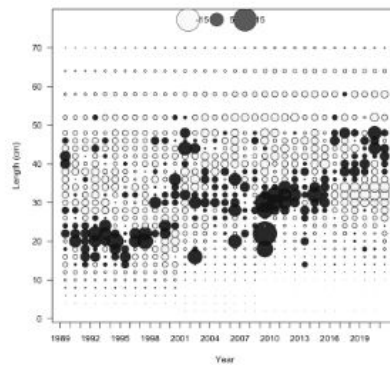
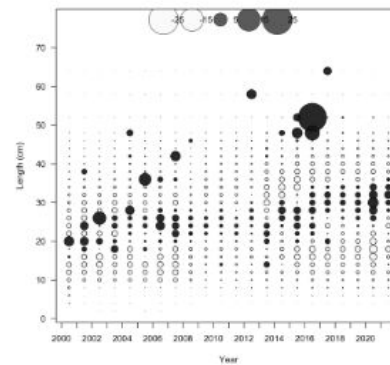


Figure 15: Pearson residuals for fits to spring recreational length composition data

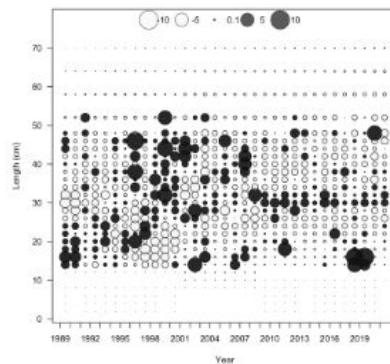
(a) Retained, N Rec Fall



(b) Discards, N Rec Fall



(c) Retained, S Rec Fall



(d) Discards, S Rec Fall

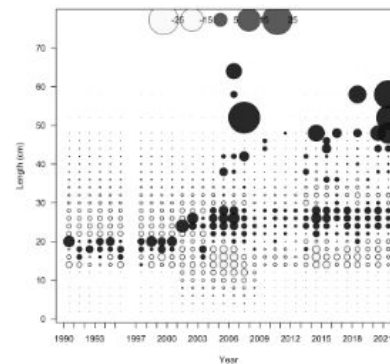
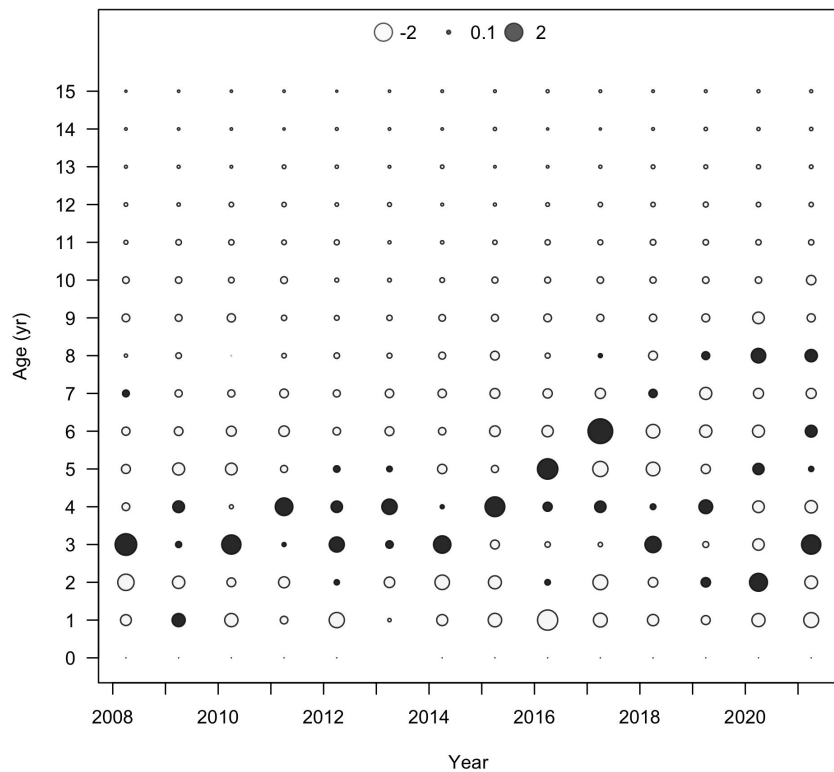
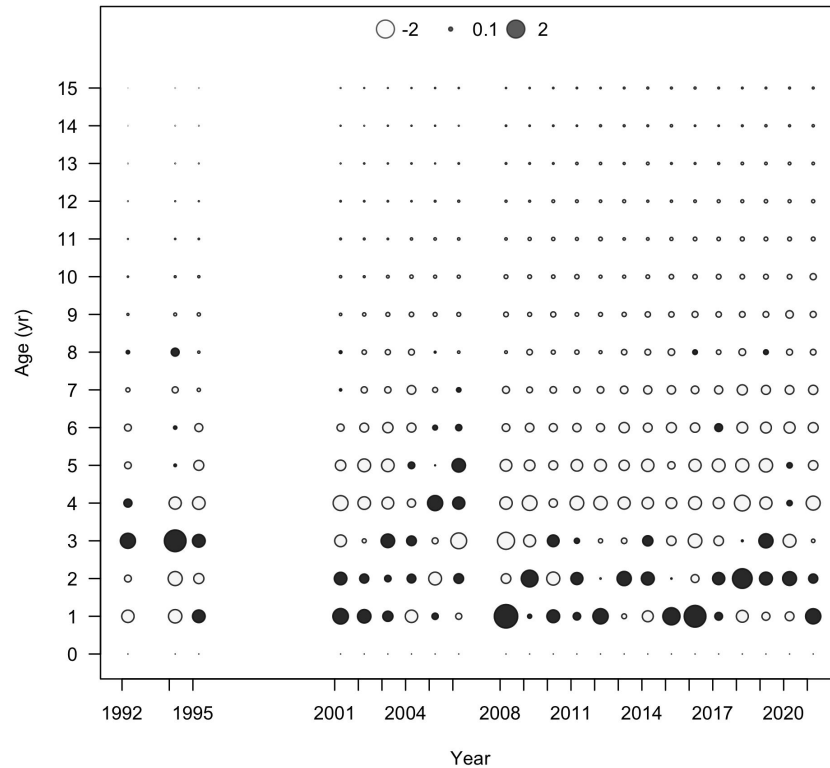


Figure 16: Pearson residuals for fits to fall recreational length composition data

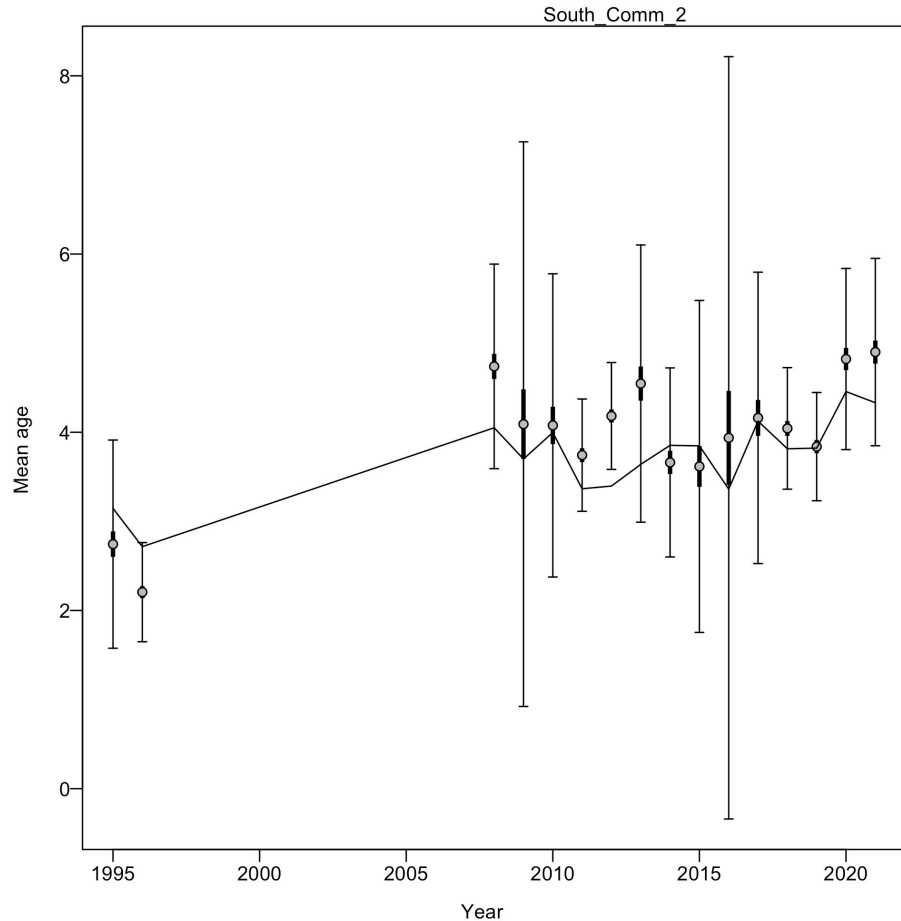
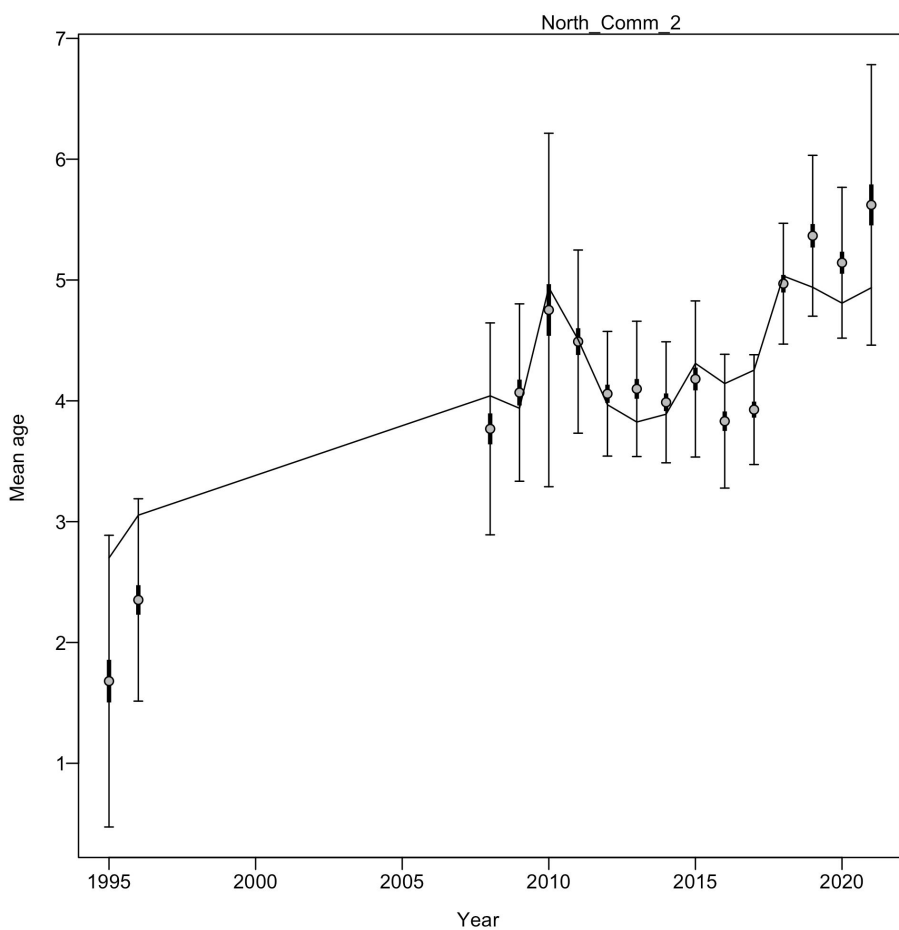
North



South

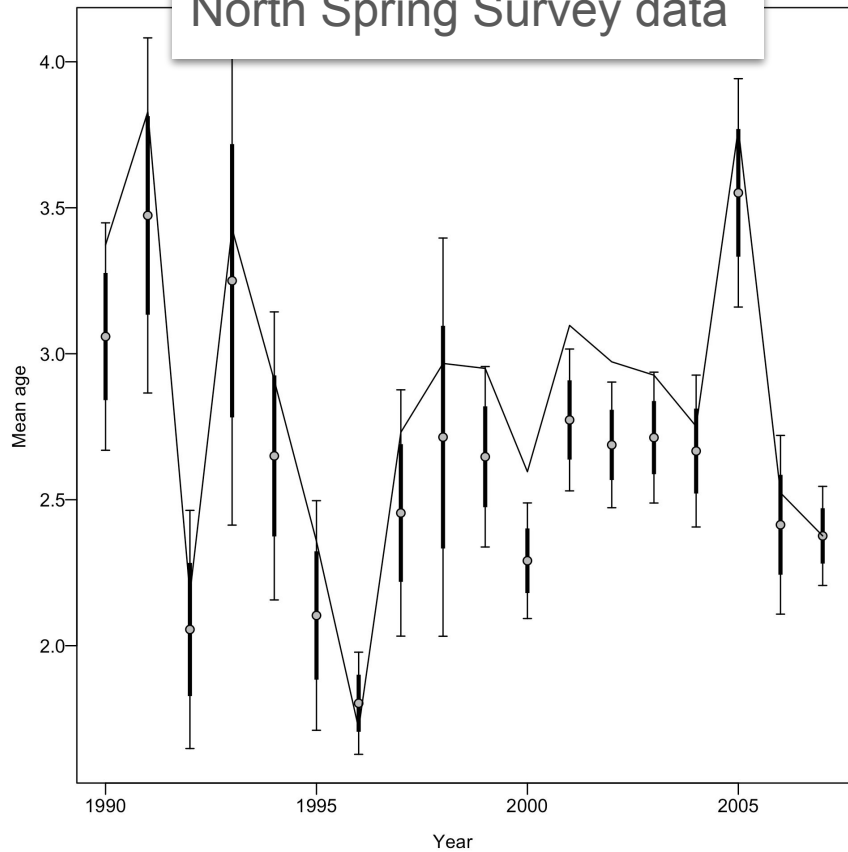


Fits to VAST age composition data

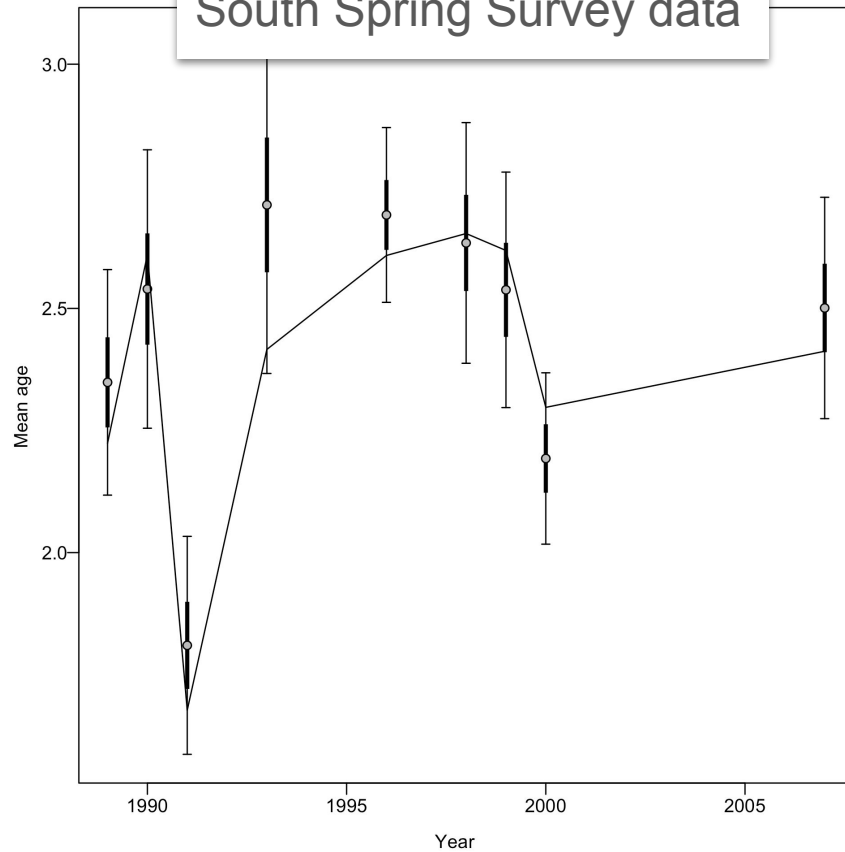


Example fits to Age at length data
(summarized by indices of mean age)

North Spring Survey data

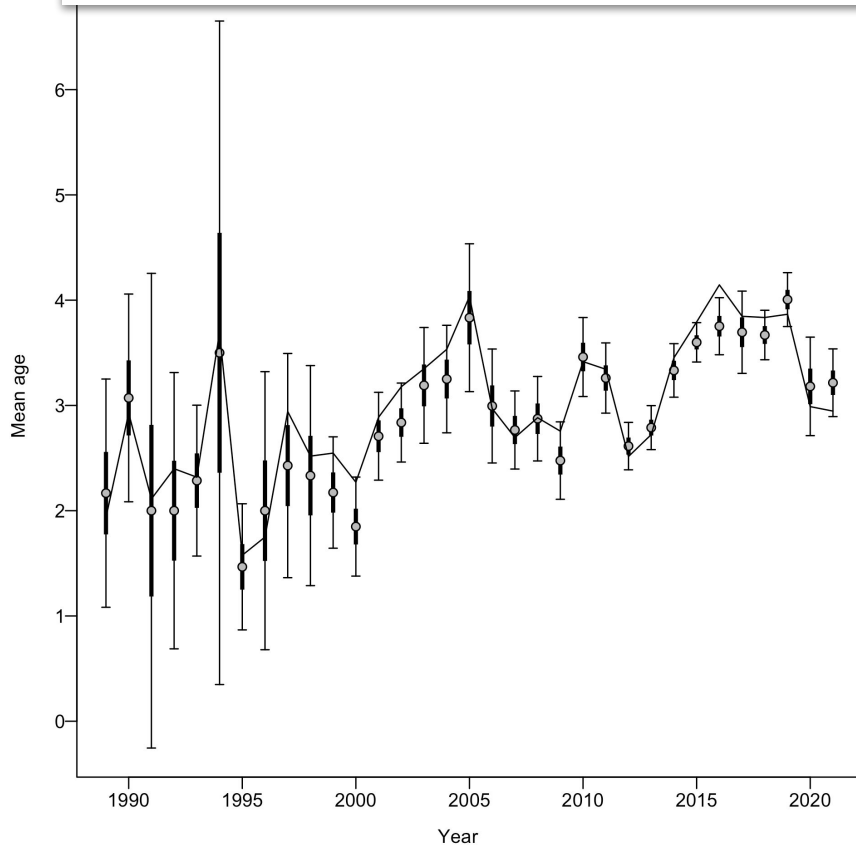


South Spring Survey data

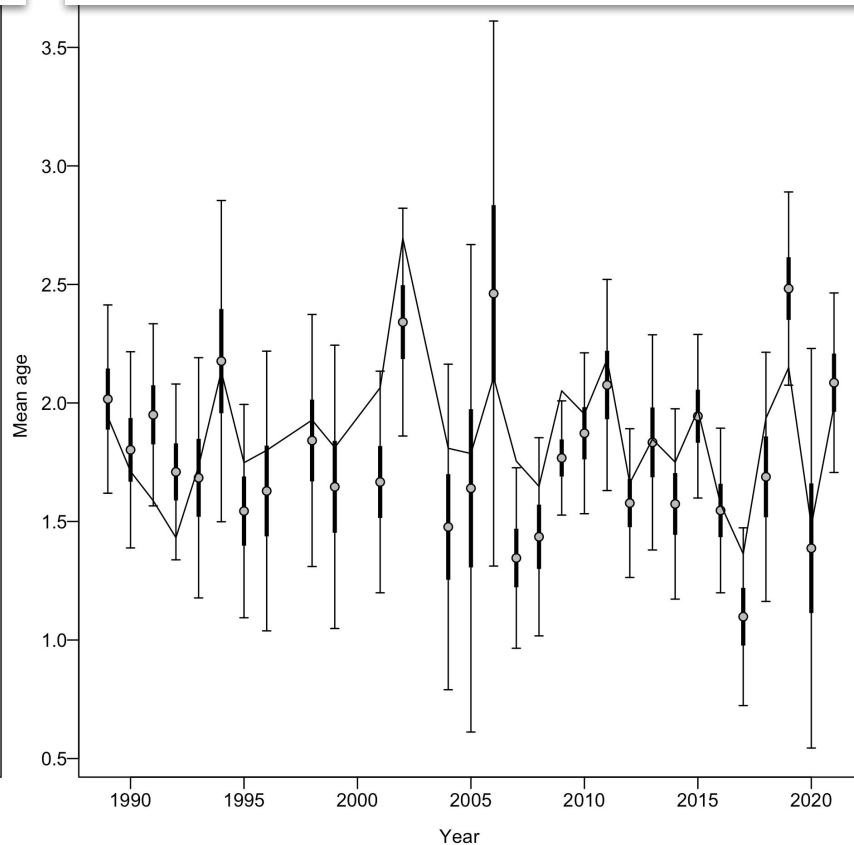


Example fits to Age at length data
(summarized by indices of mean age)

North Fall Survey age-at-length data

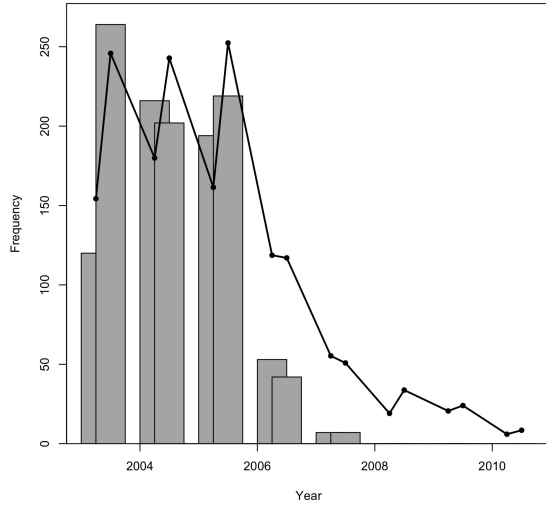


South Fall Survey age-at-length data

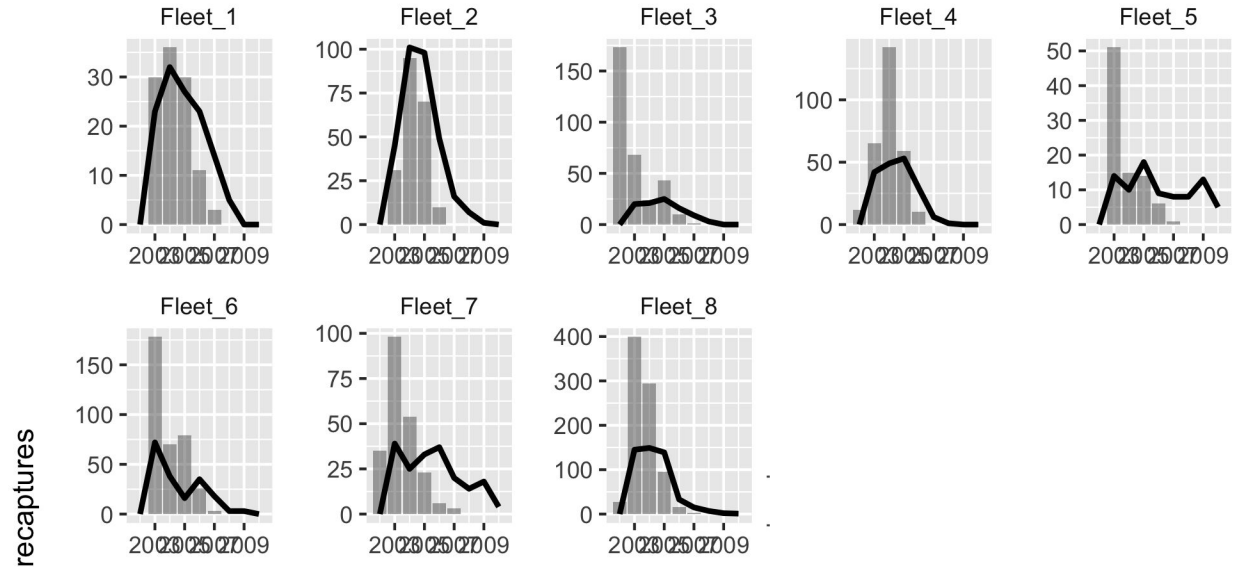


Example fits to Age at length data
(summarized by indices of mean age)

Post-latency tag recaptures aggregated across tag groups



Observed (bar) and expected (line)

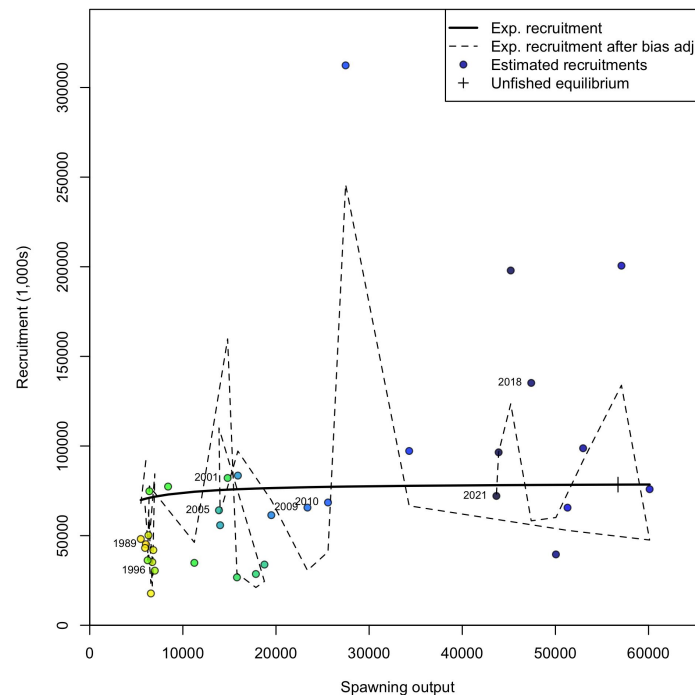
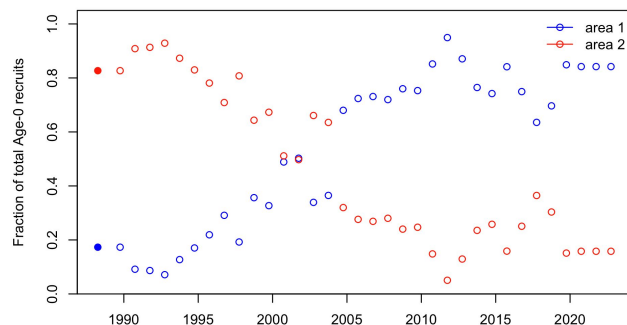
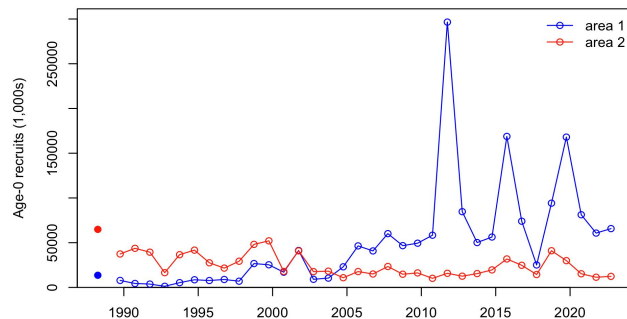
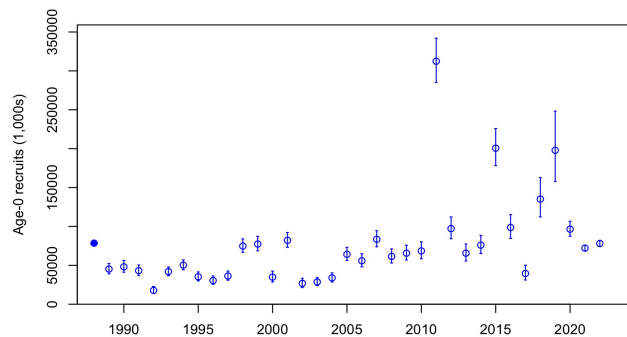


Movement

- Extent of movement estimated to be less than in 2016 SAW62 application
- 5% of northern recruited fish that are in the north move south at end of season 2

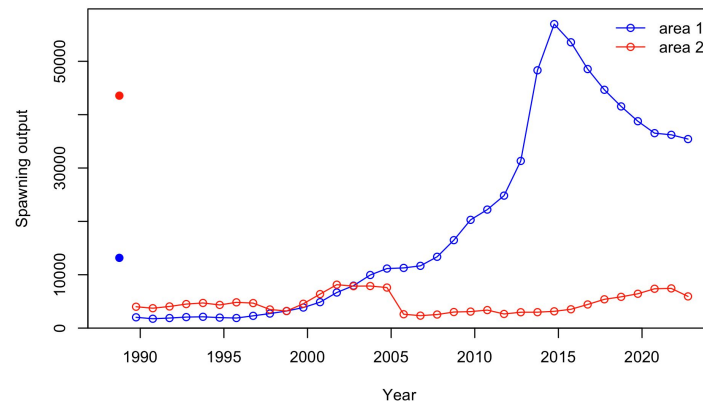
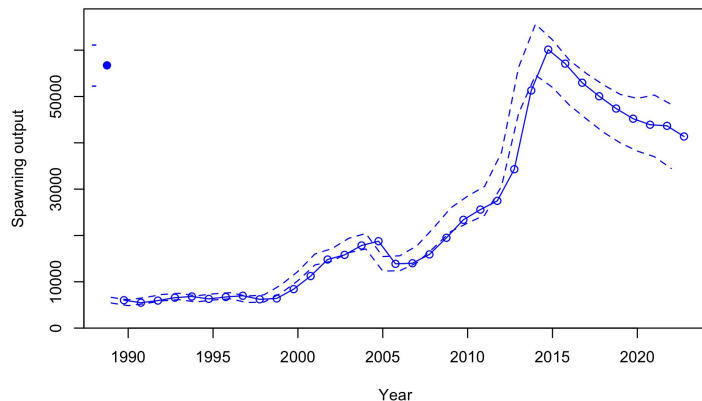
Tag recaptures

Results



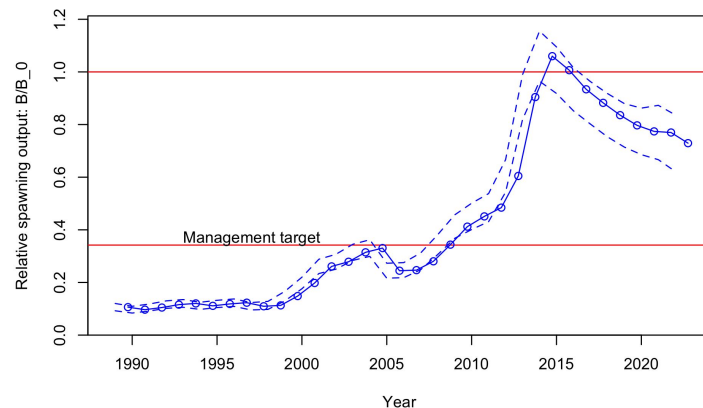
Recruitment

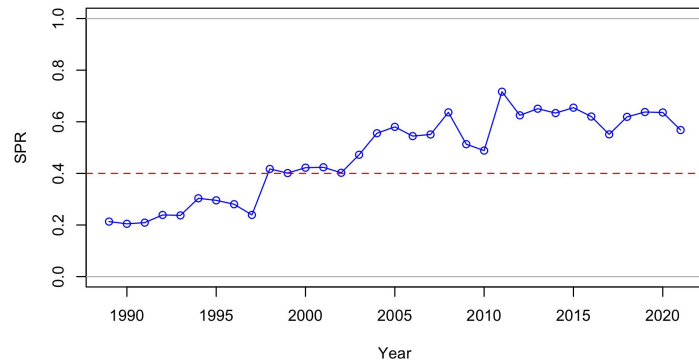
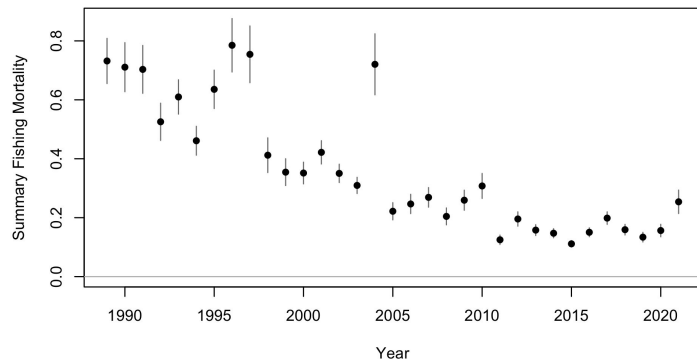
- *Increased proportion of annual recruits in the north over time*
- *Large year classes in north 2011, 2015, 2019*
- *Not capturing late 90s south recruitment*
- *bottom temperature link coefficient = 0.587*



Spawning Output

- Estimated increase in north due to 2011 year class
- Increase in South also but lower in magnitude

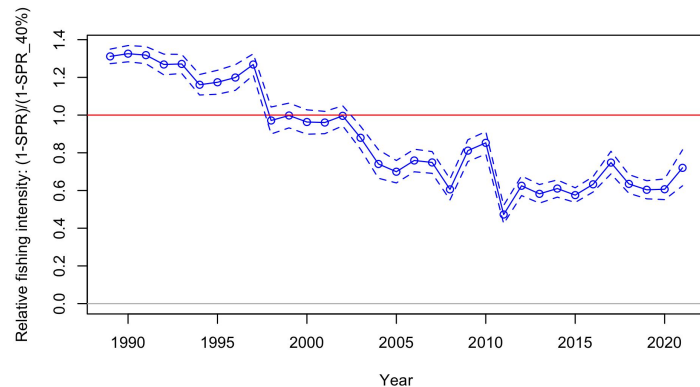


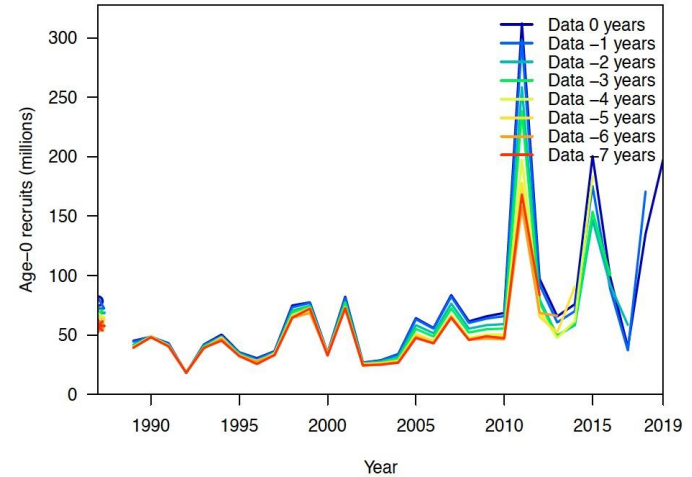
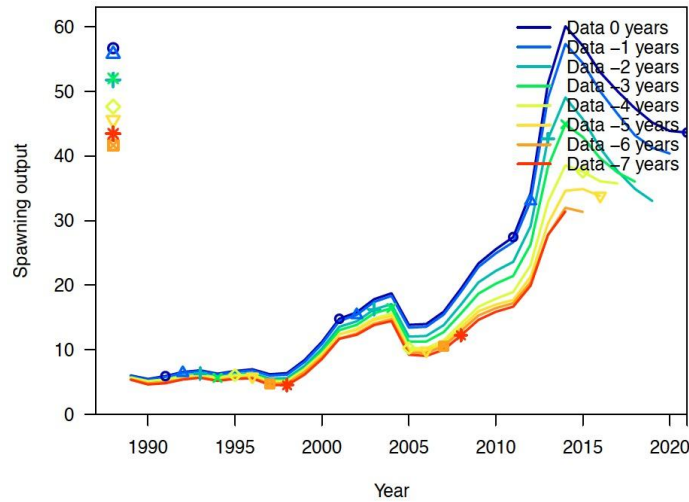


Fishing Mortality

- F averaged over ages 4-7
- Decrease in F over time, below that which reduces SPR to 40%.

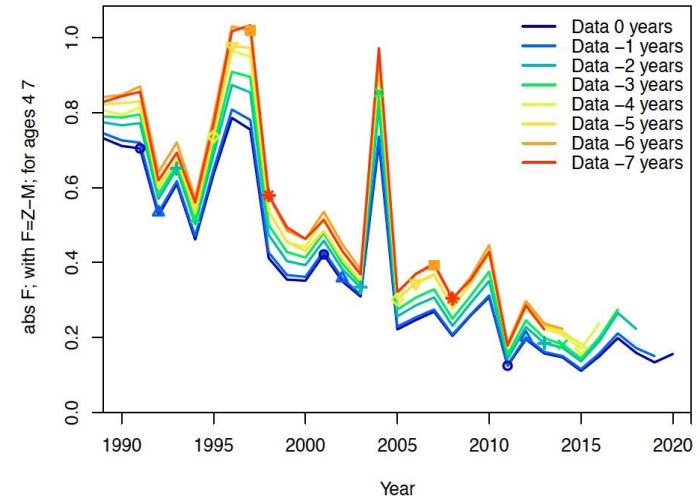
(SPR calculation uses recent 5-year average of maturity, fleet selectivity & catch pattern among fleets)

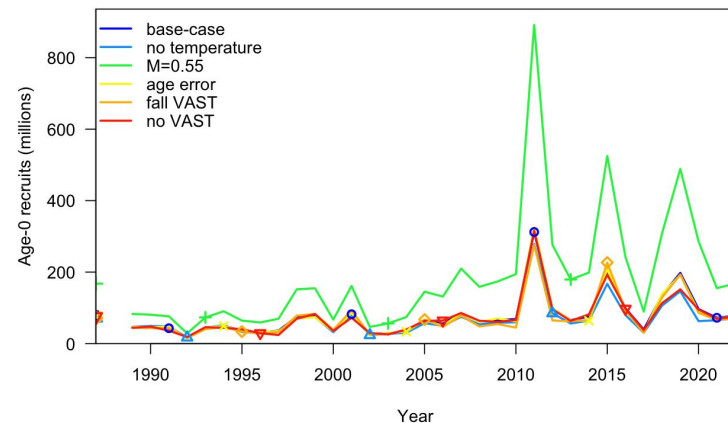
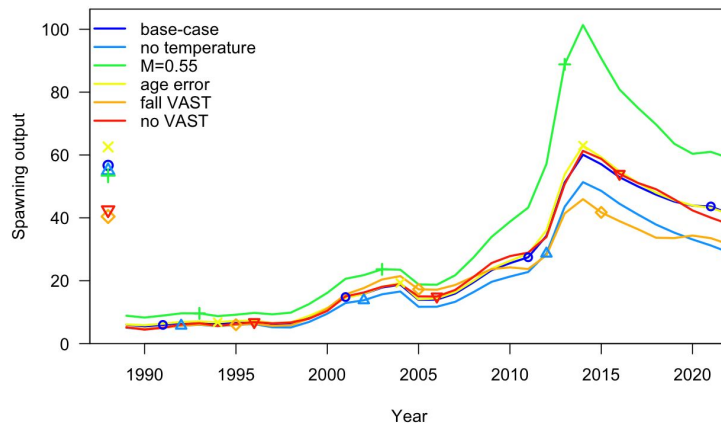




Retrospective analysis

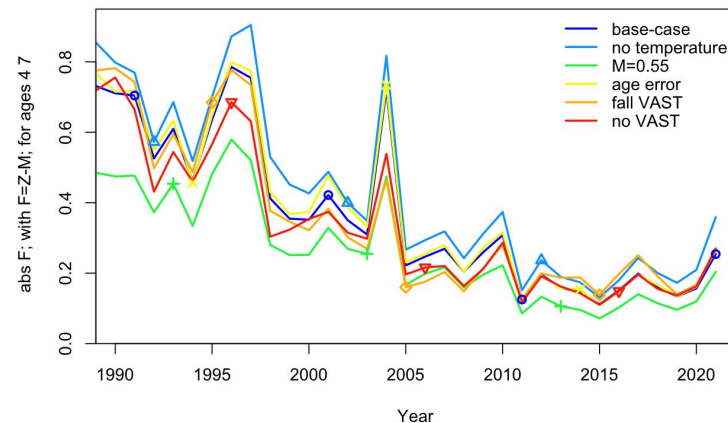
- Strong pattern in SSB and F, moderate pattern in recruitment
- Mohn's rho values:
 - SSB = -0.309, Rec = 0.104, F = 0.495





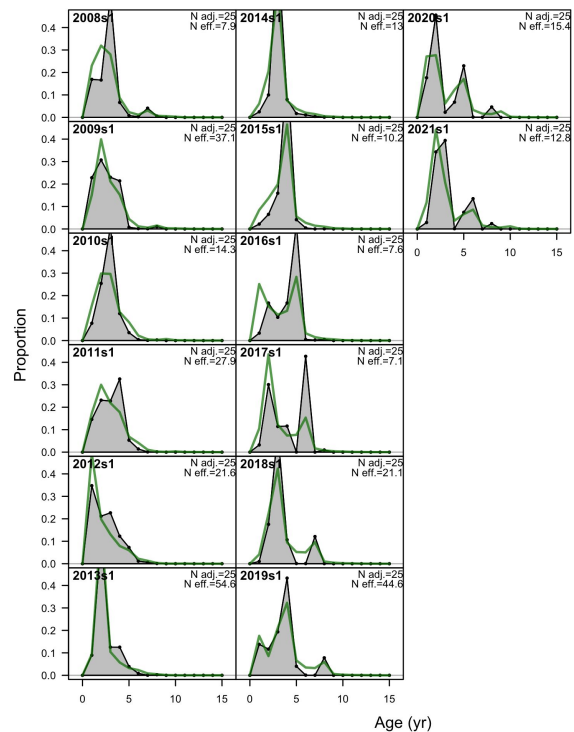
Sensitivity Analyses

- Limited number presented given SS model role
- Removing temperature link resulted in poorer model fit, decreased estimated stock size
- Fixing M to 0.55 improved overall model fit (discards, lengths, and tag recaptures). Higher estimated stock size.
- Adding ageing error improved fit to age data but little change to results
- Adding fall VAST improved fits to some data, smaller estimated stock size
- Using individual federal surveys & length comps instead of VAST indices & ages didn't fit discards as well but did improve fit to tags. Little change to results.

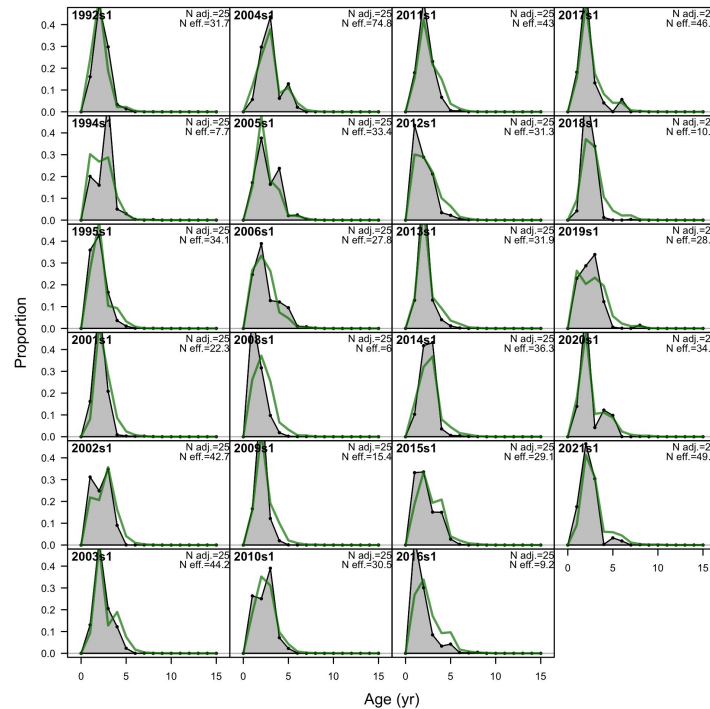


Additional slides

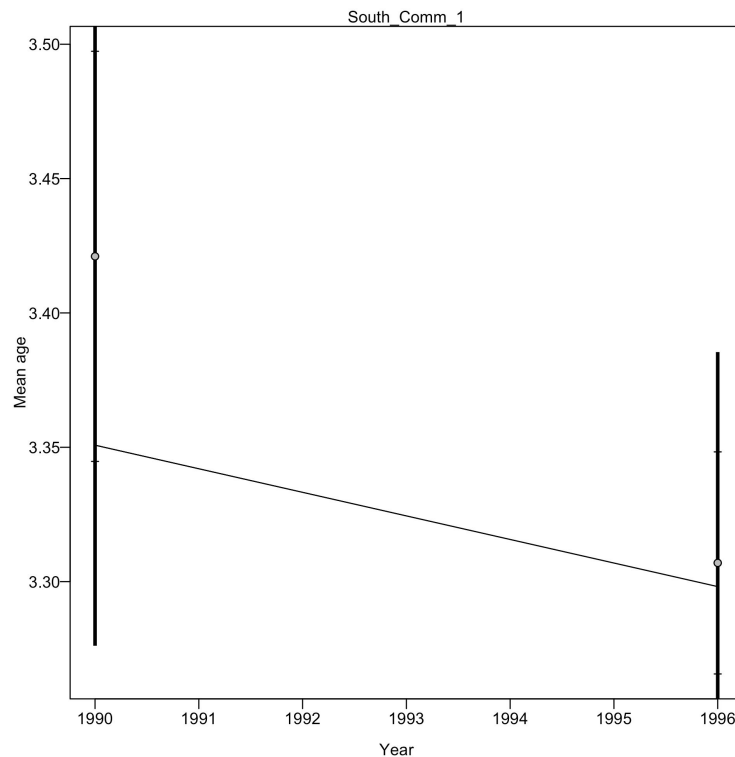
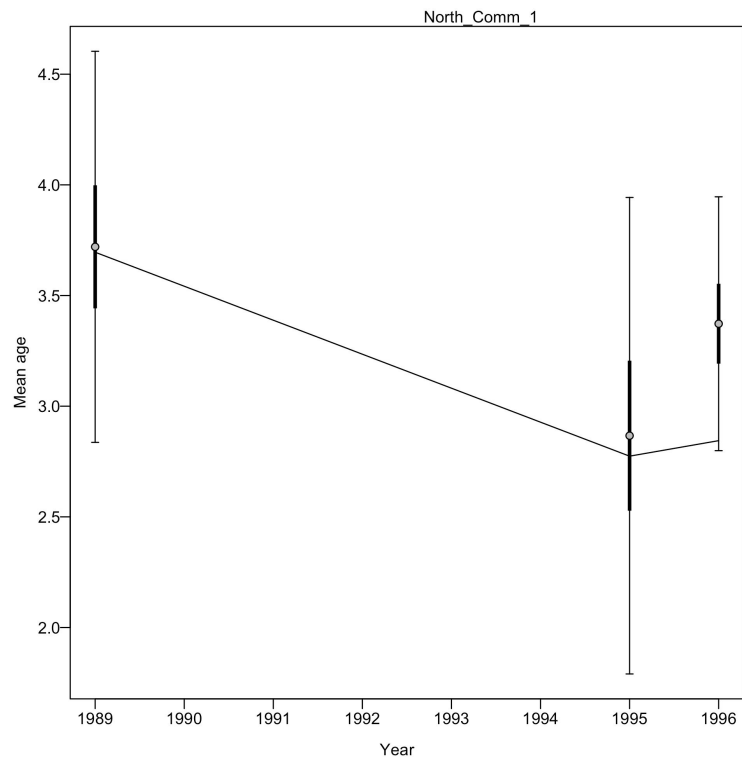
North



South



VAST age composition data



Example fits to Age at length data
(summarized by indices of mean age)